ProgressViz: An online learning platform with visual learning heatmaps, targeted feedback, and a peer challenge arena for university students.

Group 16
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1 PROBLEM STATEMENT

Online learning is now a significant part of university education, but it creates ongoing challenges for undergraduate students (Deng & Yang, 2025). In in-person classes, instructors and peers can give immediate feedback and cues on learning progress, but online learners have no clear way to tell how well they've mastered the material. This leaves them unsure about their skill growth and slowly reduces their learning motivation (Nielsen Norman Group, 2024). Student surveys also prove how harmful these gaps are: 68% of undergraduate online learners say they often "doubt their improvement" when their progress isn't clearly tracked (Nielsen Norman Group, 2024), and 52.9% rate online learning as ineffective because the progress feedback is disorganized and they lack effective guidance for understanding key concepts (Erlangga, 2022).

Existing online learning tools fail to fix this problem because they only address single pain points of effective online learning (Erlangga, 2022). Platforms like Canvas and Coursera are good at delivering course content and grading assignments, but they have no dedicated tools to help students see how well they understand core concepts. Students can only see letter grades, not which topics they've mastered, which ones confuse them, or which they need to review again. There is no existing solution that creates a system where visual progress tracking leads to targeted instructor feedback and structured peer collaboration. This traps online learners in a cycle of isolated and unguided study that hurts both their understanding of material and their learning motivation (Erlangga, 2022).

The core of these challenges lies in students losing their learning motivation, which is a problem driven by three interconnected causes: fragmented progress tracking, disorganized and delayed feedback, and social isolation. First, students often switch between platforms like Canvas, Coursera, and Zoom for different tasks during their online learning, and there's no centralized tool that maps their understanding of core concepts across tools. Without a clear view of their progress, students feel their efforts are invisible, directly reducing their motivation to persist (Nielsen Norman Group, 2024). Second, instructors can't quickly identify common confusing topics, leading to inefficient feedback, while students face inconsistent or delayed responses (Joulani et al., 2013). This lack of timely and targeted support leaves students frustrated by setbacks, further reducing their motivation to engage. Third, 72% of online learners find the experience "boring" (Erlangga, 2022), and self-directed learning becomes harder without peer interaction; students then lose the motivation to stay on track. These causes form a negative cycle: invisible progress discourages seeking feedback, which deepens isolation, and isolation ultimately kills motivation to track progress or engage with course material, trapping students in disengagement.

Thus, our design ProgressViz addresses visual progress tracking, structured feedback, and peer collaboration. Unlike existing tools that treat progress tracking, feedback, and peer interaction as separate features, ProgressViz integrates all three into a single workflow where each part drives the next. This aligns with Jonassen's (2002) key argument that online learning should center on building problem-solving skills and conceptual understanding, not just having students passively absorb content.

2 SOLUTION

To address low motivation, unclear progress, and limited support in online learning, our team designed ProgressViz, an online learning platform that combines visual progress tracking, instructor feedback, peer interaction, and personal learning management. Our concept combines visual learning journeys with peer reflection cycles for college students. Instead of treating online learning like disconnected assignments or course pages, ProgressViz helps students see how their understanding develops over time, recognize where they are struggling, and stay engaged through a clearer support system.

One of the biggest problems we identified was that students often lose motivation when they cannot tell whether they are making progress. Most online learning platforms provide materials and assignments, but they do not always help students make sense of their learning in a meaningful way. Because of that, we wanted to create something that does more than organize content. Our goal was to make progress easier to see, make confusion easier to respond to, and make online learning feel less isolating.

The core feature of our solution is the dynamic learning map, which grew out of our earlier ideas for a visual heatmap or personal growth map. This feature shows connections between course topics, completed content, and areas where students may be struggling. We chose it because many online learning systems present information in flat formats like lists or modules, which can make it hard for students to understand their learning as a whole. The map helps make growth easier to recognize and more meaningful.

Another major feature is timely, targeted feedback. Through the Quick Feedback Session, students can choose a course, check an instructor's availability, explain what they need help with, and schedule a short meeting. We included this because motivation often drops when students get stuck and do not know what to do next. A learning map can show where the problem is, but students also need a clear next step. This feature gives students a direct way to get support before frustration turns into disengagement.

We also wanted to address the isolation that often comes with online learning. ProgressViz includes a discussion space, a "compete with peers" feature, and a leaderboard with growth points, badges, and streaks. These features were designed to make learning feel more social and engaging. By giving students ways to interact with and challenge peers in a low-pressure setting, the platform encourages participation and engagement.

Another important part of ProgressViz is the dashboard and account system, which gives students one place to manage their learning. Users can access courses, check progress, manage meetings, review achievements, and track action plans. We included this because students are more likely to stay motivated when their learning environment feels organized and manageable.

Overall, ProgressViz is built around a simple idea: students are more likely to stay motivated when they can see their progress, know what to do next, get help when they need it, and feel connected to other learners.

2.1 Mockups

This section presents key mockups from our application and explains the reasoning behind the main design choices we made. These mockups were shaped by our main project goals: making progress more visible, improving access to support, and creating more opportunities for reflection and peer interaction. Combined with our user flowcharts, these interfaces illustrate how students and instructors navigate the platform and how ProgressViz's key features are interconnected.

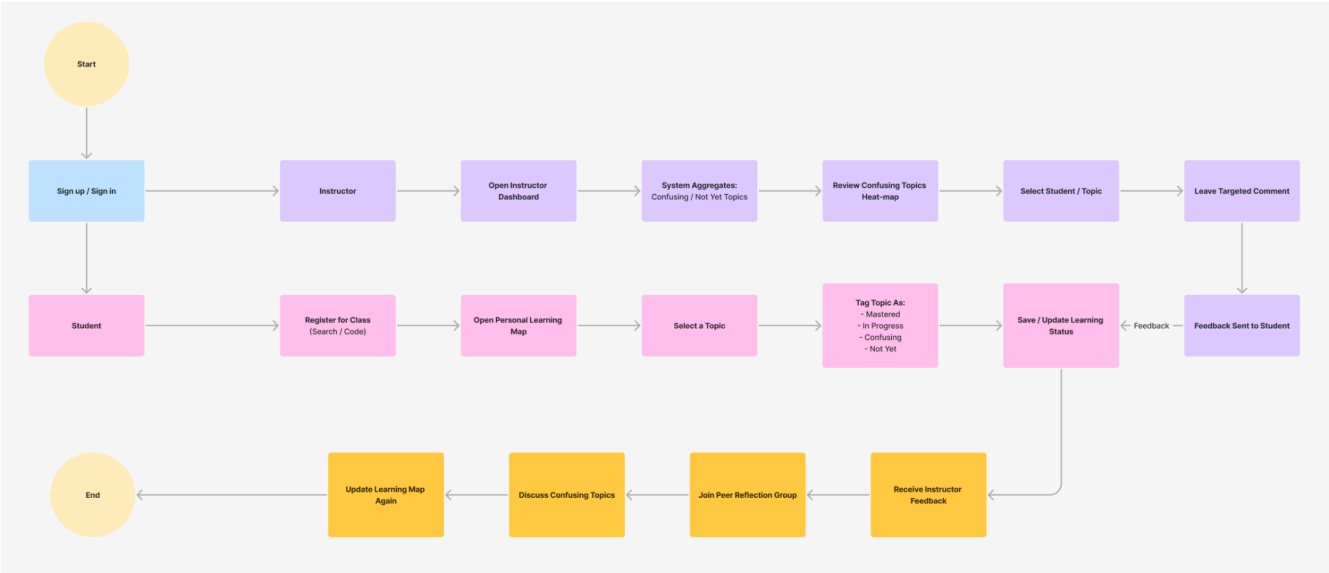

Figure 1. User Flow Map of ProgressViz for Student and Instructor Interaction

The user flow map shows how ProgressViz supports both student and instructor interaction throughout the learning process. The student side begins with sign-up, course registration, and access to the personal learning map, then continues through topic selection, progress updates, feedback, and peer reflection. The instructor side focuses on reviewing aggregated confusion data, selecting a student or topic, and leaving targeted feedback. Rather than following a one-time sequence, the flow is cyclical: students update their learning status, receive feedback, reflect with peers, and return to the learning map again. This structure reflects our goal of making online learning more continuous, interactive, and responsive over time.

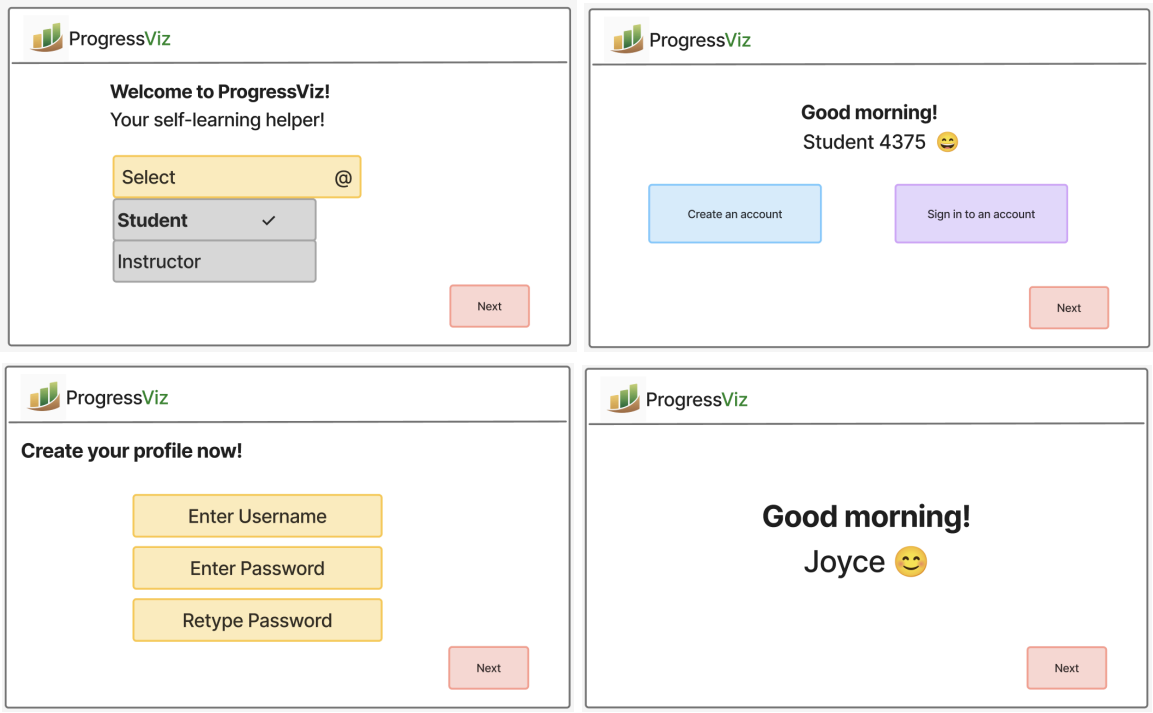

Figure 2. Log-in and Onboarding Flow

Figma Link:
https://www.figma.com/board/8YwG3fuwkDKVuI5vuZh7HF/360?node-id=0-1&p=f&t=lHZnQSs0XGBShRPX-0

Design Choice #1: Include role selection at the beginning of the onboarding process.

Rationale: We ask users to select either the “Student” or “Instructor” role at the outset so that the platform can guide them through the appropriate workflow. This helps maintain clarity and reduce confusion early in the experience. Since students and instructors use different parts of the system, role selection helps make the platform feel more personalized from the very beginning.

Design Choice #2: Use a multi-step guided flow instead of a single, cluttered entry page.

Rationale: We broke the onboarding process into multiple screens, including registration/login, account creation, and a personalized welcome screen. We chose this design because online learning platforms can be overwhelming, especially for new users. A step-by-step onboarding process makes it easier for users to adapt to the platform and helps them understand that the system is built around their personal learning journey.

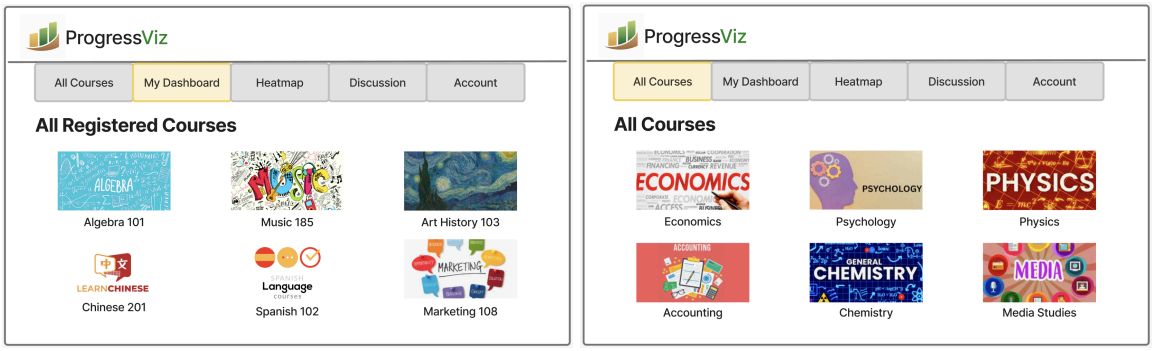

Figure 3. The “All Courses” and “My Dashboard” pages, displaying available courses and enrolled courses.

Design Choice #1: Adopt a visual card-based course layout.

Rationale: We use course cards with images and titles to make the interface easier to browse and navigate. This design helps reduce visual clutter, allowing students to quickly identify their courses. We want the platform to feel approachable rather than overwhelming, as students are more likely to stay engaged when the interface is organized and easy to understand.

Design Choice #2: Display “All Courses” and “Enrolled Courses” separately.

Rationale: We separate browsing functionality from course participation functionality because they represent different aspects of the learning experience. The “All Courses” page supports exploration and course selection, while “My Dashboard” helps students focus on the courses they are currently taking. This distinction also enhances motivation by tailoring the homepage content to the user’s personal learning context (rather than displaying generic content).

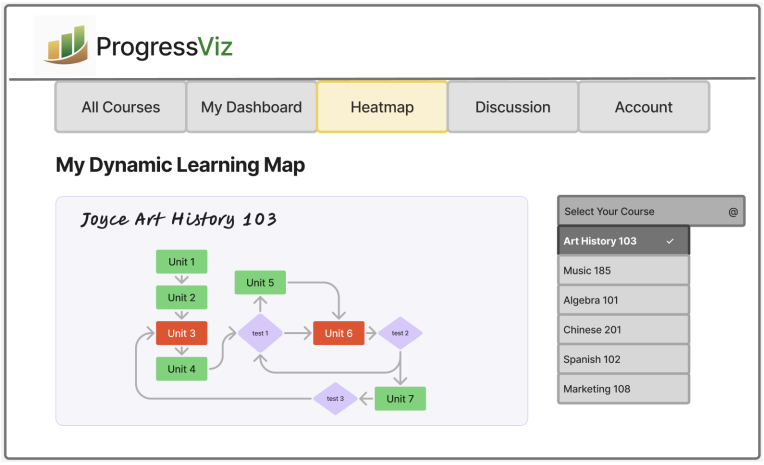

Figure 4. The Heatmap page displays a dynamic learning map for the selected course.

Design Choice #1: A node-based learning map that illustrates the relationships between units, topics, and quizzes.

Rationale: This is a core feature of ProgressViz. We designed this map to present learning as an interconnected process rather than a flat list of modules. By showing the connections between units and assessments, the map helps students understand the overall structure of the course and pinpoint their place within it. This design choice directly supports our goal of making learning progress more visible and easier to understand.

Design Choice #2: Visual progress markers to highlight completed, confusing, and upcoming content.

Rationale: We want students to quickly identify which topics they have mastered and which still require attention. This helps boost motivation, as students can more clearly see their achievements and learning gaps. Students no longer need to guess whether they are making progress; instead, they can use this map as visual evidence of their progress and areas of confusion. This design is particularly important because our project is dedicated to addressing the issue of “invisible progress” in online learning.

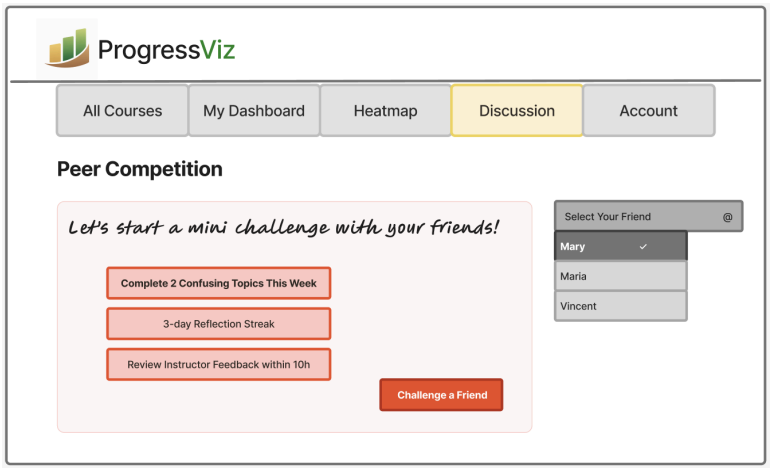

Figure 5. The discussion board homepage, which provides access to instructor feedback, peer interaction, and ranking features.

Design Choice #1: A discussion hub that integrates multiple social learning features into a single space.

Rationale: Rather than designing the discussion area as a single forum page, we have created a central hub where students can view instructor feedback, compete with peers, and check their rankings. We made this choice because we wanted discussion to feel connected to students' actual learning process rather than like a separate social feature. This also reflects our broader design philosophy: social interaction should motivate learning by helping students reflect, build connections, and stay engaged with course content.

Design Choice #2: Create separate entry points for feedback, peer competition, and rankings.

Rationale: We did not want all social activities to be jumbled together in an undifferentiated space. By providing clear, separate entry points for each feature, students can independently choose the type of support or motivation they need. Some users may require instructor guidance, while others may be more motivated by peer interaction or competitive challenges. This structure helps the platform feel more flexible and reduces the pressure on users.

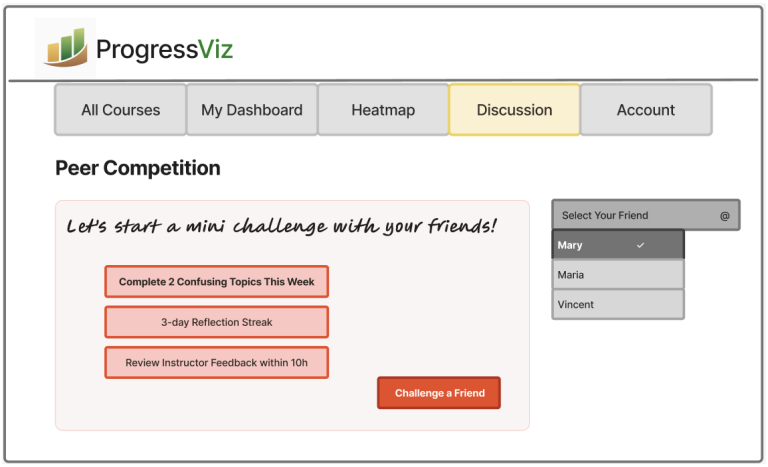

Figure 6. The "Compete with Peers" page, showcasing peer competition and the ability to issue challenges.

Design Choice #1: A low-risk peer competition feature built around small goals.

Rationale: We incorporated the "Compete with Peers" feature because online learning can feel monotonous and isolating when students study alone for extended periods. By allowing users to challenge friends, for example, to complete a challenging topic or maintain a reflection journal, the platform adds a sense of motivation and accountability. We intentionally designed the competition to feel friendly and low-pressure, so as to motivate students while avoiding unnecessary pressure.

Design Choice #2: Challenge prompts tied to actual learning behaviors.

Rationale: These challenges are not random game tasks, but are closely linked to meaningful academic behaviors, such as reviewing difficult concepts, reading instructor feedback, or maintaining a reflection journal. This is crucial because we want gamification to reinforce learning outcomes rather than distract from them. In this way, competition becomes an effective tool for supporting engagement and learning consistency.

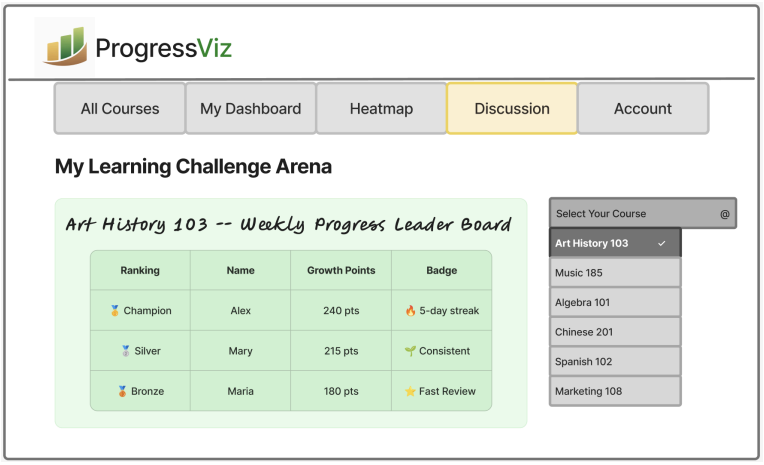

Figure 7. The Challenge Arena and Weekly Leaderboard page, displaying rankings, growth points, and badges.

Design Choice #1: A leaderboard displaying rankings, growth points, and badges for the selected course.

Rationale: We included a leaderboard because visible recognition helps sustain long-term engagement. Growth points and badges provide more nuanced and immediate signals of progress, which is particularly useful in an online learning environment where motivation often wanes between assignments or exams. This design helps students feel that their efforts are being noticed and recorded.

Design Choice #2: Implement course-specific competitions rather than a platform-wide unified ranking.

Rationale: By limiting the leaderboard to the scope of selected courses, we make the competition feel more relevant and fair. Students compare their progress within a shared academic context rather than across unrelated subjects. This also makes the feature easier to understand and establishes a more direct connection to the user's current learning activities.

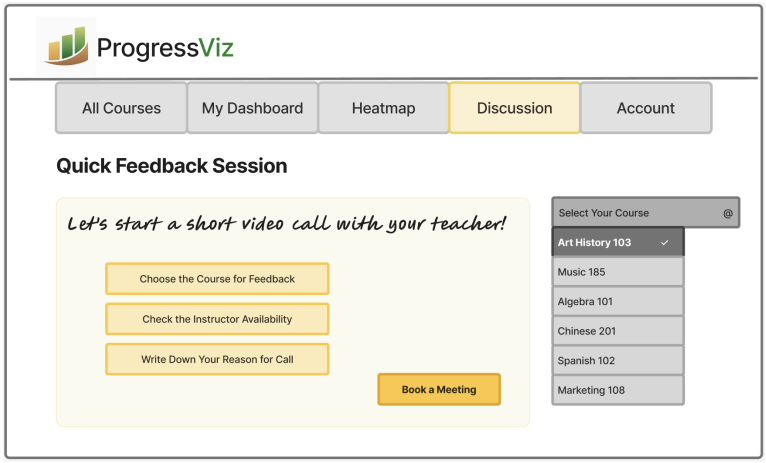

Figure 8. The "Quick Feedback Session" page, used for scheduling brief instructor support meetings.

Design Choice #1: A step-by-step booking process incorporating course selection, instructor availability, and the reason for the meeting.

Rationale: We designed this guided process because students often lose motivation when they encounter difficulties and are unsure how to proceed. This process gives them a clear next step and helps them get support before confusion turns into disengagement. Rather than leaving students to face problems in isolation, this feature empowers them to take immediate, targeted action.

Design Choice #2: Feedback requests are linked to specific courses.

Rationale: We tied meeting requests to specific courses to enable students to articulate their needs more clearly, and to allow instructors to provide more targeted responses. This also reinforces the connection between the learning map and instructor support, as students can identify specific stumbling blocks within a course and request assistance directly for that particular subject.

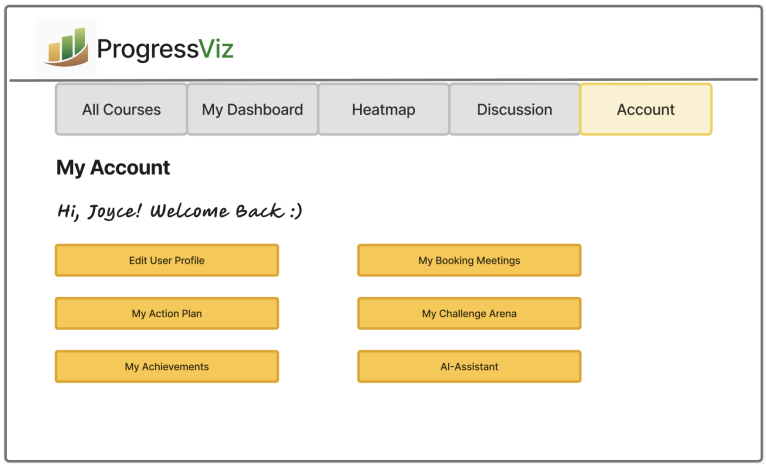

Figure 9. The My Account page, showing profile tools, bookings, action plans, achievements, challenge features, and the AI assistant.

Design Choice #1: Create a centralized account dashboard for managing various learning tools.

Rationale: We designed the account page to serve as a personal learning hub where students can manage their profile information, meeting schedules, achievement records, and various learning tools from a single, unified location. By reducing the need to switch between multiple pages, this design makes it easier for students to review goals, follow up on tasks, and track their progress. In that way, it also supports self-regulation.

Design Choice #2: Integrate Action Planning and Achievement Tracking features within the account page.

Rationale: We incorporated both the Action Plan and Achievement Log features into the design based on the premise that students are more likely to stay motivated when they can connect what they have already accomplished with what they want to do next. Specifically, the achievements feature makes learning progress more visible, while the Action Plan helps students clarify their next steps. These two features work in tandem to create a more reflective and organized learning experience for the student.

3 EVALUATION

This section explains how we tested ProgressViz to see if it actually works for students. We looked for mistakes using expert rules and then checked our improvements with real users. This process helped us make sure the final product really helps students manage their classes without getting confused.

3.1 Heuristic Evaluation

During the middle of our project, our team checked our design against a list of 10 standard rules of thumb for good design, called heuristics. We realized that while we wanted the app to look clean and simple, we had accidentally made it harder for users to see what was happening or stay in control.

When we looked at the Dynamic Learning Map, we found a big problem with the Visibility of System Status. In the original design, every unit box looked exactly the same. This is a problem because the system doesn't show the user what they have finished and what they are doing right now. Because there were no progress bars or icons, users had to use their own memory to find their place. This made the learning process feel broken and frustrating, which is the opposite of what we wanted to do.

We also found a problem with User Control and Freedom on the Quick Feedback booking page. There were no back or cancel buttons on the form. If a student clicked the page by mistake, they were "trapped" and had to refresh the whole browser to leave. This creates a stressful experience and makes students less likely to use the tool again when they actually need help from an instructor.

Finally, we noticed that the design was not consistent. The My Account page used completely different colors and button styles compared to the rest of the app. This sudden change makes the app feel like it was built in separate, disconnected pieces. It lowers the user's trust and makes the platform feel less professional and more annoying to use.

3.2 User Tests

After the heuristic evaluation, we conducted user tests with our target audience to see how real students interact with ProgressViz. Since our initial design was already based on student needs, the tests mostly confirmed that our core ideas were effective. Most students found the simple layout easy to use and helpful for organizing their studies.

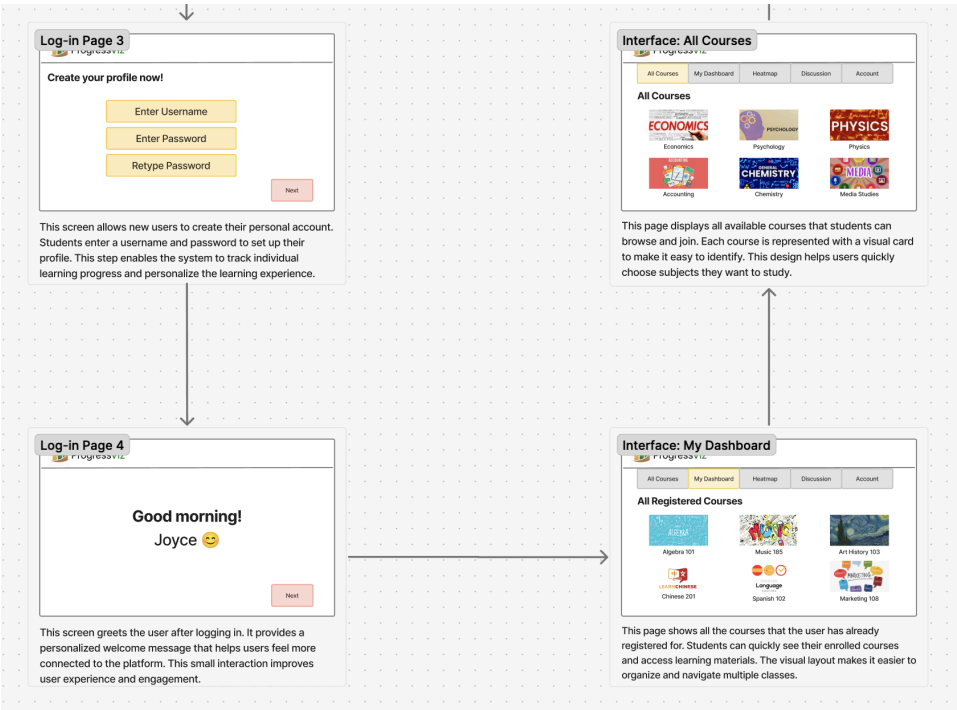

Figure 10. The overall user flow of ProgressViz from account creation to learning management.

During the tests, the Heatmap received very positive feedback. Students liked how it made their learning progress visible and easier to understand. However, the tests confirmed our earlier finding that small visual cues, like color changes for finished units, are necessary to keep users motivated. By using bright green for finished work, we ensure that students can see their growth at a single glance without any confusion.

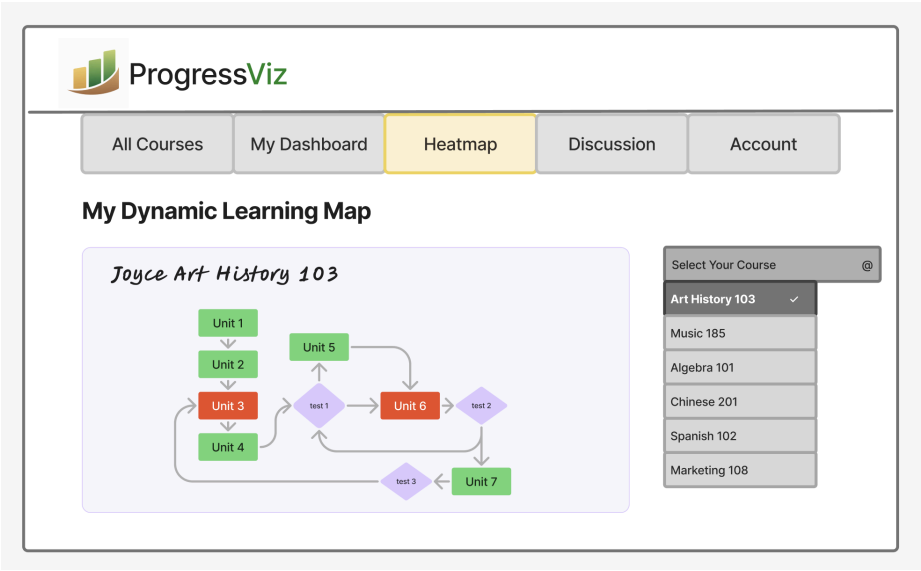

Figure 11. The core Heatmap feature that helps students track unit connections and progress.

We also tested the Discussion Board features, including the Quick Feedback Session and Peer Competition. Students noted that having a fast way to book a meeting with an instructor is a key feature for handling difficult topics. While the overall structure worked well, the testing reminded us that keeping the Cancel or Back buttons visible is vital so that students always feel in control of the booking process.

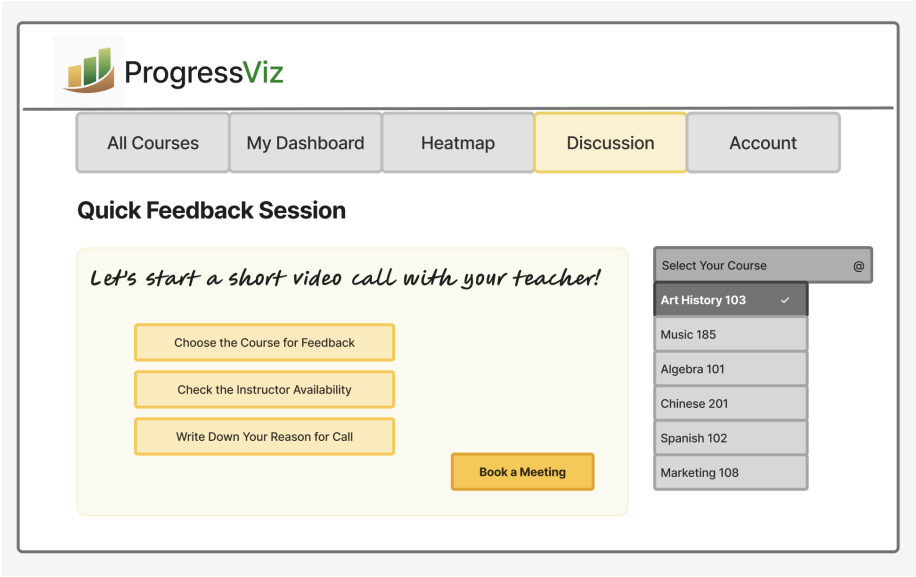

Figure 12. The booking interface designed for timely support and easy navigation.

In summary, the user tests showed that our final design successfully balances a clean look with helpful tools. Students felt that the app centralized their personal learning information well, making it much easier to track their activities across different classes.

4 LIMITATION

Our design relies heavily on consistent user and community engagement. The Dynamic Learning Map only provides value if students update their progress daily; without regular input, the visual benefits disappear. Furthermore, social features like Peer Competition and Quick Feedback depend on others being active. If classmates do not join challenges or instructors do not respond quickly, the user may feel discouraged and stop using the platform.

ProgressViz currently lacks full support for users with disabilities. Because the system relies significantly on color coding and complex visual diagrams, students with visual impairments or color blindness may find navigation difficult. We assumed all users could interact with a highly visual interface. Without alternative modes like text to speech or high contrast settings, some students may be excluded from the experience.

In our pursuit of a minimalist clean look, we accepted trade-offs that sometimes hinder navigation. Our evaluation showed that removing extra buttons, such as Back or Cancel, to keep the UI simple actually made users feel trapped. While a minimalist design helps with focus, it can make complex tasks harder by removing clear emergency exits. We chose simplicity over flexibility, which may not align with every user's navigation preference.

Finally, the design assumes every student has reliable mobile access and a stable internet connection. As a web based tool, it is ineffective in offline environments. Additionally, ProgressViz focuses exclusively on academic paths. It does not support broader social connections or non-academic activities, narrowing the app’s scope to a very specific type of student interaction.

5 APPENDIX

Appendix A – Heuristic Evaluation

Appendix B – User Test Observations

Appendix C – Figma Design File

6 REFERENCES

Deng, Z., & Yang, Z. (2025). Exploring the impact of online education on student engagement in higher education in post-COVID-19: what students want to get? *Frontiers in Psychology*, *16*, 1574886. https://doi.org/10.3389/fpsyg.2025.1574886

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Erlangga, D. T. (2022). Student problems in online learning: Solutions to keep education going on. *Journal of English Language Teaching and Learning (JELTL)*, *3*(1), 21–26.

Jonassen, D. H. (2002). Engaging and supporting problem solving in online learning. *The Quarterly Review of Distance Education*, *3*(1), 1–13.

For Extra Credit:

Code:
https://github.com/Joyce061026/info360group16/blob/main/index.html

Interactive Prototype:
https://joyce061026.github.io/info360group16/

Username: Joyce
Password: 123456
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